Module_3

Team Members:

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Project Title

Medical X-Ray Images Pneumonia Prediction of Cancer

Project Goal

The project aims to explore Kaggle's pediatric pneumonia chest X-ray dataset and build a deep learning model to classify pneumonia from pediatric chest X-ray images. The model is analyzed if there is a correlation/trend between the pathways that upregulate tumor growth/angiogenesis, and the organ in which the tumor develops.

Disease Background:

Cancer is defined by two hallmarks of cancer and genes you are interested in researching based on that decision. Then fill out the information below.

1. Cancer hallmark focus: Angiogenesis
2. Genes of interest: VEGFA, VEGFB, VEGFC, FLT1, KDR

Angiogenesis is the process by which cancer cells induce nearby blood vessels form to feed growing tumor. Tumors need oxygen and nutrients to grow beyond a very small size (about 1-2 mm). To achieve this, cancer cells release pro-angiogenic factors — most importantly VEGF (Vascular Endothelial Growth Factor) — which stimulates nearby normal blood vessels to grow new branches that grow into the tumor. Angiogenesis genes are cancer-cancer types, and are associated with a number of genomic and transcriptomic features. Further, it has noted that macrophages and Treg infiltration were generally associated with angiogenesis-related gene expression in most cancer types, suggesting that immune cells may contribute to tumor angiogenesis in multiple cancer types.

AKA you're focusing on a single cancer type or looking across cancer? Specifically at your decision, guide the section to include relevant information about the disease at the appropriate level of detail. Regardless, each tumor cells should be filled in. If you are looking at multiple cancer types, you should investigate differences between the types (e.g. what is the most prevalent cancer type? What is the most aggressive cancer type? Which cancer type has the most/least mutations?). Also, what is the most/least common cancer driver mutation? Which cancer type is associated with the most/least immune infiltration? Note that this is a variable data that will be used in the analysis.

Limiting intends at long cancer (using this data: https://gdc.cancer.gov/files/public/file/TCGA-CDR-SupplementalTableS1.xlsx+&+https://doi.org/10.1016/j.cell.2018.02.052#supplementary-2)

- Prostatectomy (Prostate)

1,116 Million new cases and 19 million deaths, lung cancer was the second most commonly diagnosed cancer and the leading cause of cancer death in 2020.This cancer is broadly classified into two main types: non-small cell lung cancer (NSCLC) and small cell lung cancer (SCLC). NSCLC accounts for about 85% of lung cancers and the 5-year survival (SCLC, which is the most aggressive type, exactly for about 10% cancer.

- Bile Ducts (gender: Male & Female)

Bile Ducts include cholangio, and gallbladder, and some certain gland(s) (Gallbladder).

- Bladder (of urothelium)

Despite the actual amount of numerous therapeutic modalities, such as surgery, radiation therapy, chemotherapy, immunotherapy, and targeted therapy, the tum rate decreasing from treatment to treatment. Bladder cancer is a heterogeneous disease with different molecular characteristics, clinical presentations and a wide health concern.

- Biliary Cancer (Gender: Male & Female)

Angiogenesis effects in lung cancer patients: tumor growth enhancement, metastasis promotion, changes in immunological microenvironments. Substantial Growth Factors (EGF) Although each protein is the VEGF family is a unique biological function, they all interact with EGFR to activate the downstream pathways that drive cell proliferation, angiogenesis, and invasion

Data Set:

Once you decide on the subject of data you want to use, you write "cancer type + name, key clinical features, location" which genes will you look at?) describe the dataset. This is a list of clinical features, or you don't need to describe them at all, only the ones planned to your question.

Describe the data set and all metadata. Did the successful of the data set. Describe how the data was collected — What techniques were used? What units are the data measured in? Etc.)

We will be looking at the RNA Sequencing data and the genes of VEGF (Vascular Endothelial Growth Factor) cells as an indicator for angiogenesis and comparing that to multiple cancer models leading to the cancer. The data set we are using is the TCGA Pan-Cancer (PanCan) Atlas https://xena.ucsc.edu/public-datasets/TCGA-CDR-SupplementalTableS1.xlsx+&+https://doi.org/10.1016/j.cell.2018.02.052#supplementary-2 TCGA, and compared to other factors. We will model with structure algorithms and data concentrations to indicate tumor growth.

Data Analysis:

Methods

The machine learning technique we are using is Principal Component Analysis (PCA). PCA finds high-dimensional gene-expression data and decomposes it into new features that find the directions of greatest variance in the dataset. In our case, PCA helps us visualize 20 VEGFRAM Genes pathway activity across thousands of samples and determine if tumors cluster or spread in ways that correlate with cancer type or clinical features.

What is the method optimized? What does the model decide to do (good/bad)? PCA maximizes variance while minimizing the projection error. It's considered "good enough" when the first principal components explain a meaningful amount of variance and the plot shows clear structure or patterns.

Analysis

Describe how you analyzed the data. This is where you should interpret your PyCharm code so that anyone reading this can run your code to perform the analysis that you decided, you figures, etc.)

To analyze the data, we used the TCGA dataset provided in class. First, we loaded the gene expression file and the metadata file, which include cancer type information. We then selected several cancer genes of interest (TP53, EGFR, KRAS, PIK3CA, and BRAF) and combined their expression values with the corresponding cancer type labels. After removing any missing data, we applied Principal Component Analysis (PCA) to reduce the high-dimensional gene expression data to two principal components. The purpose was to visualize if the genes that matched the pathway were VEGFA, VEGFB, VEGFC, FLT1, KDR, FLT4, FLT3, FLT3LG, and VEGFC1. We filtered the dataset to include only these genes and the samples from the cancer types of interest. Then, we combined the gene data with the metadata to assign samples back to gene. The PCA was then performed on the gene expression matrix, and the first two principal components were used to visualize how samples from different cancer types cluster in gene expression space. This approach helps us see if tumors from the same cancer type have similar gene expression patterns and if different cancers show distinct molecular signatures. Next, we plotted PC1 versus PC2 for the selected cancer types, using different colors for each cancer type. We created the gene data which

Verify and validate your analysis:

We are clustering our data and running a silhouette analysis on a PCA plot to test the significance of the clustering. We clustered between two cancer types, k=2, that showed visual distinct clustering, shown through a scatter plot. To test this visual hypothesis, we calculated the silhouette score, which was 0.5746, shown with a silhouette plot. This deeems that these two cancer types, KIRC and LIHC, clusters are reasonably distinct. PC1 explains 61.5% of overall variation in this dataset. By looking at the PCA loadings, we were able to find what genes had the most influence in PC1 variation. This was displayed in a bar graph.

Conclusions and Ethical Implications:

Based on our silhouette score on clustering these two groups, s = 0.5746, these two cancer types have distinct clustering. Our PCA loading for PC1 identifies NRP1, NRP2, KDR, VEGFA, and IL1B to be the top 5 genes contributing to this variation. We are looking into VEGFA because it's an indicator for angiogenesis VEGFA has a loading score of 0.359719 contributing to PCA1 variance. That means there's a moderate correlation between this gene, VEGFA, and the principle component. Bias interpretation and misinterpreting this data can lead to a misdiagnosis. PCA's also compress datasets, so this reductoin could marginalize underrepresented biological groups.

Limitations and Future Work:

We wonder how the gene upregulation and variation contributes to all cancer types. With these trends, can there be targeted therapies for these pathways to inhibit angiogenesis and vascularization? Future work would include testing this method on multiple data sets to see if this is a continous pattern across many data sets and marginalized biological groups. For future work, we would want to continue to test our method and look more into the direct affect on angiogenesis. There could be a clearer correlation between VEGFA gene and the distinct cancer clusters.

NOTES FROM YOUR TEAM:

What an interesting project!

QUESTIONS FOR YOUR TA:

Thank you for all your help!